A VQE Study of the Lipkin Interaction

FYS5419 – Quantum computing and quantum machine learning

PROJECT 1

Egil Furnes
egilsf@uio.no
github.com/egil10/fys5419

University of Oslo
Department of Physics
March 11, 2026

Abstract

The aim of this project is to study the so-called Lipkin model, which is a simplified Hamiltonian, using the VQE algorithm.

1 Introduction

Quantum computing offers a route to simulating quantum many-body systems in a way that is difficult to reproduce with classical resources, an idea often traced to early arguments by Feynman (1982) about using controllable quantum systems to emulate nature. In parallel, the modern field of quantum machine learning (QML) studies how parameterized quantum circuits can be used for inference, optimization, and simulation tasks, often within hybrid quantum–classical workflows suitable for noisy intermediate-scale quantum (NISQ) devices (Schuld and Petruccione 2018; Schuld and Killoran 2021; Carleo et al. 2019).

A central example of such hybrid methods is the Variational Quantum Eigensolver (VQE), which approximates the ground state of a quantum Hamiltonian using a parameterized quantum circuit evaluated on a quantum processor and optimized by a classical algorithm (Peruzzo et al. 2014; McClean et al. 2016). Variational circuit methods are closely related to other NISQ-era algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) (Farhi et al. 2014).

The aim of this project is to study a simplified interacting many-body Hamiltonian, known as the Lipkin model, and to compare classical diagonalization with VQE-based energy estimation across a sequence of increasingly structured test problems. We begin by constructing the computational basis for one and two qubits and studying the action of elementary operators and gates, including the Pauli matrices, Hadamard gate, phase gate, and the controlled-NOT (CNOT) gate. This provides a controlled setting for generating entangled states such as the Bell states and for quantifying bipartite entanglement through reduced density matrices and the von Neumann entropy.

We then analyze a simple two-level Hamiltonian written both as a matrix representation and in terms of Pauli operators, and we study how its eigenvalues and eigenvectors evolve as an interaction parameter is varied. This serves as a baseline for implementing the VQE algorithm and comparing its results with exact classical solutions.

Finally, we extend the analysis to interacting two-qubit systems and to the Lipkin Hamiltonian, originally introduced as a solvable model for testing many-body approximation methods (Lipkin et al. 1965). The Hamiltonian is rewritten in terms of Pauli operators so that expectation values can be evaluated using quantum circuits. Classical diagonalization results are then compared with VQE simulations, allowing us to study how interaction strength influences spectral properties, state mixing, and entanglement.

2 Methods

2.1 Initial definitions

First, we introduce some basic components as part of QML. One-qubit bases $|0\rangle$ and $|1\rangle$ are defined in eq. (1).

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{1}$$

Using these one-qubit bases, we can write quantum states as a linear combination as shown in eq. (2), or using the Bloch-sphere representation rewritten as in eq. (3)

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \tag{2}$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\phi}\,|1\rangle \tag{3}$$

Quantum algorithms are typically formulated using Pauli operators $\sigma_x, \sigma_y, \sigma_z$ as seen in eq. (4) and a small universal set of quantum gates that act on qubits. The Pauli matrices together with the identity $\mathbf{I}$, form a convenient operator basis for expressing quantum Hamiltonians and observables.

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4}$$

In practical implementations, these operators are realized through elementary gates such as the Hadamard gate (H), which creates superposition, the phase gate (S), which introduces controlled phase shifts, and the CNOT gate,

which generates entanglement between qubits. All as seen in eq. (5). These gates form a standard set of building blocks for constructing quantum circuits and implementing algorithms such as the VQE.

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \quad S = \begin{pmatrix}1 & 0\\ 0 & -i\end{pmatrix}$$
$$\text{CNOT} = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix} \tag{5}$$

2.2 Tensor products and two-qubit states

When describing systems of multiple qubits, the Hilbert space of the composite system is constructed using tensor (Kronecker) products of the individual qubit states as seen in eq. (6). This allows basis states of multi-qubit systems to be written as products of the single-qubit computational basis states $|0\rangle$ and $|1\rangle$. For a two-qubit system the computational basis therefore consists of the four states $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, obtained through tensor products of the single-qubit states.

$$\begin{aligned} |00\rangle &= |0\rangle \otimes |0\rangle \quad |01\rangle = |0\rangle \otimes |1\rangle \\ |10\rangle &= |1\rangle \otimes |0\rangle \quad |11\rangle = |1\rangle \otimes |1\rangle \end{aligned} \tag{6}$$

Certain superpositions of these two-qubit basis states form maximally entangled states known as Bell states as seen in eq. (7). These states play an important role in quantum information theory and serve as fundamental examples of quantum entanglement between two qubits.

$$\begin{aligned} |\Phi^+\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \\ |\Psi^+\rangle &= \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \end{aligned} \tag{7}$$

For composite quantum systems, operators acting on multiple qubits are constructed using tensor products of single-qubit operators. If an operator A acts on the first qubit and an operator B acts on the second qubit, the combined operator acting on the two-qubit system is written as $A \otimes B$. In this way, Hamiltonians and quantum gates for multi-qubit systems can be expressed in terms of tensor products of Pauli matrices and the identity operator.

As an example, the operator $I \otimes \sigma_x$ flips the second qubit while leaving the first qubit unchanged. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ this operator is given in eq. (8).

$$I \otimes \sigma_x = \begin{pmatrix}0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix}. \tag{8}$$

Similarly, tensor products between Pauli operators act on both qubits simultaneously. For example, the operator $\sigma_x \otimes \sigma_x$ flips both qubits as seen in eq. (9).

$$\sigma_x \otimes \sigma_x = \begin{pmatrix}0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0\end{pmatrix}. \tag{9}$$

While pure quantum states are written as state vectors, a more general description is given by density matrices. For a pure state $|\psi\rangle$ the density operator is given in eq. (10) and for a statistical ensemble $\{|\psi_i\rangle, p_i\}$ it becomes as in eq. (11).

$$\rho = |\psi\rangle\langle\psi|, \tag{10}$$

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|. \tag{11}$$

For bipartite systems the state of a subsystem is obtained via the partial trace, in eq. (12), which is central when analysing entanglement. A standard measure of quantum uncertainty is the von Neumann entropy in eq. (13), which vanishes for pure states and is positive for mixed states.

$$\rho_A = \text{Tr}_B(\rho_{AB}), \tag{12}$$

$$\mathcal{S}(\rho) = -\text{Tr}(\rho \log \rho), \tag{13}$$

2.3 Eigenvalue analysis

The stationary Schrödinger equation defines the energy spectrum of a quantum system. The Hamiltonian operator acting on a quantum state produces the same state scaled by its corresponding energy eigenvalue, as seen in eq. (14), where $\mathcal{H}$ is the Hamiltonian and E the energy eigenvalue and $|\psi\rangle$ the corresponding eigenstate.

$$\mathcal{H}|\psi\rangle = E|\psi\rangle \tag{14}$$

In the computational basis the Hamiltonian can be represented as a matrix acting on the state vector, and for a two-level system this can take the form in eq. (15). Here the diagonal elements correspond to energies of basis states while the off-diagonal elements represents interactions between them.

$$\mathcal{H} = \begin{bmatrix}\mathcal{H}_{11} & \mathcal{H}_{12}\\ \mathcal{H}_{21} & \mathcal{H}_{22}\end{bmatrix} \tag{15}$$

Furthermore, the Hamiltonian is decomposed into $\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_1$, with a solvable non-interacting component $\mathcal{H}_0$ and an interaction term $\mathcal{H}_1$. See eq. (16), where E_i represent the unperturbed energy levels and V_{ij} describe interaction matrix elements.

$$\mathcal{H}_0 = \begin{bmatrix}E_1 & 0\\ 0 & E_2\end{bmatrix} \quad \mathcal{H}_1 = \begin{bmatrix}V_{11} & V_{12}\\ V_{21} & V_{22}\end{bmatrix} \tag{16}$$

Any 2×2 Hamiltonian can be expressed using the Pauli operator basis together with the identity operator. For the

present system this leads to the identities seen in eq. (17) with eq. (18).

$$\begin{aligned}\mathcal{H} &= a\mathbf{I} + b\sigma_x + c\sigma_y + d\sigma_z \\ \mathcal{H}_0 &= \mathcal{E}\mathbf{I} + \Omega\sigma_z \\ \mathcal{H}_1 &= c\mathbf{I} + \omega_z\sigma_z + \omega_x\sigma_x\end{aligned} \tag{17}$$

$$\mathcal{E} = \frac{E_1 + E_2}{2} \quad \Omega = \frac{E_1 - E_2}{2} \tag{18}$$

Now, we can move on to the analytical eigenvalue solution, where the eigenvalues of the Hamiltonian are obtained from the characteristic equation in eq. (19).

$$\det(\mathcal{H} - E\mathbf{I}) = 0 \tag{19}$$

For a general 2×2 Hamiltonian in eq. (20) the resulting energy eigenvalues can be found in eq. (21). Using this, the lowest eigenvalue corresponds to the ground state energy of the system.

$$\mathcal{H} = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \tag{20}$$

$$E_\pm = \frac{a+d}{2} \pm \sqrt{\left(\frac{a-d}{2}\right)^2 + b^2} \tag{21}$$

To study the effects of interactions in eq. (22), the Hamiltonian is parametrised by a coupling parameter $\lambda \in [0, 1]$, where the limits $\lambda = 0$ and $\lambda = 1$ correspond to the non-interacting and fully interacting systems respectively.

$$\mathcal{H} = \mathcal{H}_0 + \lambda\mathcal{H}_1 \tag{22}$$

The Hamiltonian is diagonalised numerically to obtain its eigenvalues and eigenvectors. For Hermitian matrices this is efficiently performed using the `numpy` package in `python`, as seen in listing 1.

```
np.linalg.eigh(H)
```

Listing 1. Diagonalization of the Hamiltonian matrix $\mathcal{H}$ using NumPy's `np.linalg.eigh`, which computes the eigenvalues and eigenvectors of a Hermitian operator. This routine is used as the classical baseline for validating the quantum variational results.

2.4 Variational Quantum Eigensolver (VQE)

VQE is based on the variational principle, for any normalised trial state $|\psi(\theta)\rangle$, the expectation value of the Hamiltonian provides an upper bound to the true ground state energy. The goal in using a VQE is therefore to minimize the energy expectation value ewith respect to the circuit parameter θ shown in eq. (23).

$$E_0 \leq \min_\theta \langle\psi(\theta)| \mathcal{H} |\psi(\theta)\rangle \tag{23}$$

A parameterized quantum circuit $U(\theta)$ prepares a trial state from a simple initial computational basis state. By varying the parameters θ, the circuit explores different candidate states in eq. (24)

$$|\psi(\theta)\rangle = U(\theta)\,|0\rangle^{\otimes n} \tag{24}$$

To evaluate the energy on a quantum computer, the Hamiltonian is expressed as a weighted sum of tensor products of Pauli operators, in eq. (25). Each term can then be measured seperately on the quantum circuit, where c_k are the real coefficients and P_k are Pauli strings acting on the qubits.

$$\mathcal{H} = \sum_k c_k P_k \quad P_k \in \{I, X, Y, Z\}^{\otimes n} \tag{25}$$

The VQE cost function is the expectation value of the Hamiltonian with respect to the trial state. Using the Pauli decomposition, the energy can be written as a sum of expectation values of Pauli operators in eq. (26).

$$E(\theta) = \langle\psi(\theta)| \mathcal{H} |\psi(\theta)\rangle = \sum_k c_k \langle P_k\rangle_\theta \tag{26}$$

A classical optimizer updates the circuit parameter iteratively in order to reduce the energy expectation value. Conceptually, this can be written as an update rule, where η is a learning rate or step size, as in eq. (27).

$$\theta_{t+1} = \theta_t - \eta\nabla_\theta E(\theta_t) \tag{27}$$

After convergence, the optimal parameters θ^* define the approximate ground state prepared by the circuit. The estimated ground state energy is then shown in eq. (28).

$$E_{\text{VQE}} = E(\theta) \tag{28}$$

The choice of parameterized circuit $U(\theta)$ determines the family of states explored by the VQE and therefore strongly influences the quality of the variational solution. For a single qubit, a minimal ansatz consists of a single rotation gate is seen in eq. (29).

$$|\psi(\theta)\rangle = R_y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle\,. \tag{29}$$

For two qubits, a common hardware-efficient architecture combines single-qubit rotations with an entangling gate. A simple example is shown in eq. (30).

$$|\psi(\boldsymbol{\theta})\rangle = \big(R_y(\theta_3)\otimes R_y(\theta_4)\big)\,\text{CNOT}\,\big(R_y(\theta_1)\otimes R_y(\theta_2)\big)\,|00\rangle\,. \tag{30}$$

2.5 The Lipkin model

The Lipkin–Meshkov–Glick (LMG) model describes N fermions distributed over two N-fold degenerate energy levels separated by an energy ε. In terms of quasispin operators the Hamiltonian is written as in eq. (31).

$$\mathcal{H} = \varepsilon J_z + \frac{1}{2}V(J_+^2 + J_-^2) \tag{31}$$

The operators J_z, J_+, J_- satisfy the $\mathfrak{su}(2)$ algebra and act within the angular momentum sector $J = N/2$. The first term represents the single-particle energy splitting, while the interaction term scatters particle pairs between the two levels. To simulate the model on a quantum computer, the quasispin operators are mapped to Pauli operators acting on N qubits. This yields the qubit Hamiltonian shown in eq. (32).

$$\mathcal{H} = \frac{\varepsilon}{2}\sum_{i=1}^{N} Z_i + \frac{V}{2}\sum_{i<j}(X_iX_j - Y_iY_j) \tag{32}$$

This Pauli representation allows the Lipkin Hamiltonian to be directly evaluated within the VQE framework introduced in the following subsection.

2.6 Implementation in `python`

All simulations and numerical verification routines were implemented in `python`. Linear algebra and numerical routines were handled using `numpy` and `scipy`, data handling with `pandas`, and visualization with `matplotlib` and `seaborn`. Quantum circuits and hybrid quantum–classical optimization were implemented using `PennyLane` and `Qiskit` (NumPy Developers 2024; Xanadu AI 2024; IBM Quantum 2024). Python has become a standard platform for numerical computing and quantum algorithm prototyping due to its extensive scientific ecosystem.

The full implementation and supporting code are available in the project repository (Furnes 2026), with the development environment summarized in listing 2 in the Appendix. Parts of the implementation follow examples and structures from the quantum machine learning teaching repository by Hjorth-Jensen et al. (2026).

2.7 Use of AI-assisted development tools

AI-assisted programming tools were used during development for code generation, debugging, and documentation. These included large language models such as `ChatGPT` (OpenAI 2023), `Claude` (Anthropic 2026), `Grok` (xAI 2026), and `Gemini` (Google DeepMind 2026), as well as coding assistants such as `GitHub Copilot`, `Antigravity` and `Cursor` (GitHub 2026; Google 2026; Anysphere 2026).

3 Results

3.1 Single-Qubit Operations and Bell State Entanglement

Applying the Pauli operators defined in section 2 confirms the expected behaviour.

$$\sigma_x|0\rangle = |1\rangle \qquad \sigma_x|1\rangle = |0\rangle \qquad \text{bit-flip} \tag{33}$$

$$\sigma_y|0\rangle = i|1\rangle \qquad \sigma_y|1\rangle = -i|0\rangle \qquad \text{bit-flip + phase} \tag{34}$$

$$\sigma_z|0\rangle = |0\rangle \qquad \sigma_z|1\rangle = -|1\rangle \qquad \text{phase-flip} \tag{35}$$

The Hadamard gate (H) maps computational basis states to equal superpositions, $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$. The phase gate introduces a relative phase of i on $|1\rangle$ while leaving $|0\rangle$ unchanged. In our results, both gates act as expected.

Starting from $|00\rangle$, the circuit $H \otimes \mathbf{I}$ followed by CNOT produced eq. (36), and with repeated measurement of $|\Phi^+\rangle$ with 50 runs times 1000 shots yielded the table seen in table 1.

$$|00\rangle \xrightarrow{H\otimes\mathbf{I}} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle \tag{36}$$

Outcome	μ	σ
$\|00\rangle$	0.4988	0.018
$\|01\rangle$	0.0000	0.000
$\|10\rangle$	0.0000	0.000
$\|11\rangle$	0.5012	0.018

Table 1. Mean measurement probabilities (μ) and standard deviations (σ) obtained from repeated circuit evaluations.

The outcomes are exclusively $|00\rangle$ and $|11\rangle$ with equal probability, which is the exact expected signature of the $|\Phi^+\rangle$ Bell state. The two qubits are perfectly correlated, measuring one immediately determines the other. The statistical fluctuations of $\sigma = \pm 0.018$ are consistent with the expected noise, as in eq. (37)

$$\sqrt{\frac{p(1-p)}{N_{\text{shots}}}} \approx 0.016 \tag{37}$$

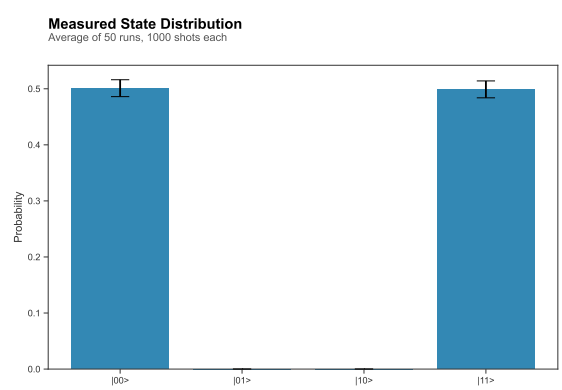


Figure 1. Measurement Distribution

Computing the von Neumann entropy $\mathcal{S} = -\mathrm{Tr}\,(\rho_A \log_2 \rho_A)$ in table 2.

State	Type	S
$\|\Phi^+\rangle$	Maximally entangled	1.000
$\|\Phi^-\rangle$	Maximally entangled	1.000
$\|\Psi^+\rangle$	Maximally entangled	1.000
$\|\Psi^-\rangle$	Maximally entangled	1.000
$\|00\rangle$	Product state	0.000
$\|+\rangle \otimes \|0\rangle$	Product (superposition)	0.000

Table 2. Von Neumann entropy S for several two-qubit states. The Bell states $\left|\Phi^{\pm}\right\rangle$ and $\left|\Psi^{\pm}\right\rangle$ are maximally entangled with $S = 1$, while product states have vanishing entropy.

3.2 Two-Level System: Exact Diagonalization vs. VQE

Using parameters $E_1 = 0, E_2 = 4, V_{11} = 3, V_{22} = -3, V_{12} = V_{21} = 0.2$ the Pauli decompositions are shown in table 3.

Coefficient	Value	Expression
$\mathcal{E}$	2.0	$(E_1 + E_2)/2$
Ω	-2.0	$(E_1 - E_2)/2$
c	0.0	$(V_{11} + V_{22})/2$
ω_z	3.0	$(V_{11} - V_{22})/2$
ω_x	0.2	V_{12}

Table 3. Pauli decomposition coefficients obtained from the project parameters $E_1 = 0$, $E_2 = 4$, $V_{11} = 3$, $V_{22} = -3$, and $V_{12} = V_{21} = 0.2$.

The Eigenvalues $E_{\pm}(\lambda)$ are computed both analytically and numerically (using `numpy.linalg.eigh`) over 101 values of $\lambda \in [0, 1]$, where the maximum discrepency between the two methods are $\approx 10^{-15}$, as shown in fig. 2.

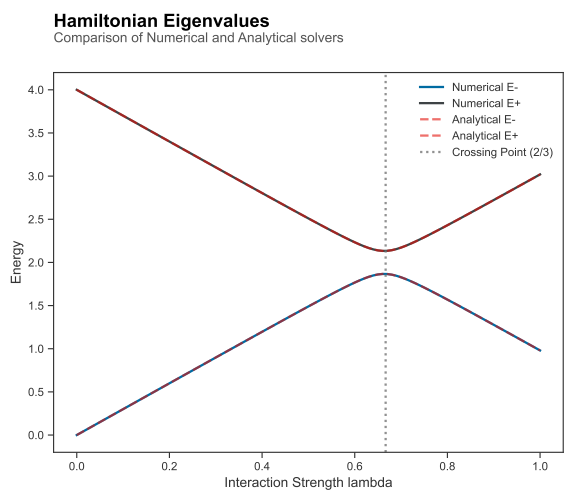


Figure 2. Eigenvalues

The spectrum shows and avoided crossing at $\lambda = \frac{2}{3}$, where are this point, the diagonal matrix elements cross $(3\lambda = 4 - 3\lambda)$, but the off-diagnoal coupling $V_{12} = 0.2$ prevents a true degeneracy, creating a gap of $\Delta E \approx 0.27$.

Tracking the eigenvector decomposition of the ground state as a function of λ, as seen in table 4.

λ	Prob($\|0\rangle$)	Prob($\|1\rangle$)	Character
0.0	100%	0%	Pure $\|0\rangle$
$\frac{2}{3}$	$\sim$50%	$\sim$50%	Maximally mixed
1.0	$\sim$1%	$\sim$99%	Dominated by $\|1\rangle$

Table 4. Measurement probabilities of the computational basis states $|0\rangle$ and $|1\rangle$ as a function of the interaction parameter λ.

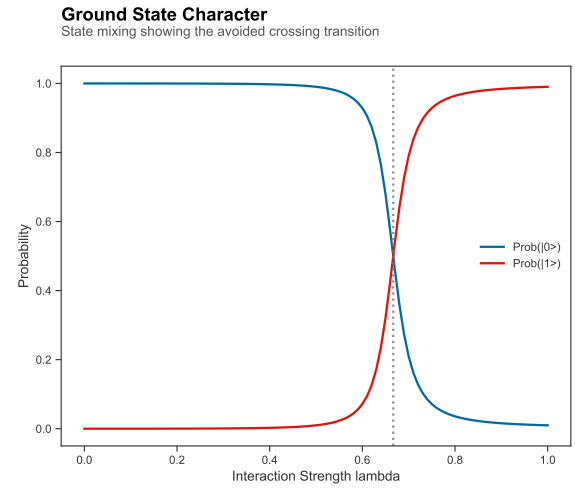


Figure 3. Eigenvector Mixing

The ground state character inverts as λ passes through the crossing point. Below $\lambda = \frac{2}{3}$ it is predominantly $|0\rangle$, and above, $|1\rangle$ dominates. This transitionis smooth because the coupling V_{12} mixes the two basis states. At $\lambda = 1$ only $\approx 1\%$ of $|0\rangle$ remains, which is consistent with the theoretical prediction.

The trial state is shown in eq. (38), where the single-parameter rotation spans the full real single-qubit Hilbert space, since the Hamiltonian is real and symmetric, R_y is exactly sufficient.

$$|\psi(\theta)\rangle = R_y(\theta)\,|0\rangle = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)|1\rangle \tag{38}$$

The energy is evaluated as $E(\theta) = c_0 + c_z\langle\sigma_z\rangle_\theta + c_x\langle\sigma_x\rangle_\theta$ where $\langle\sigma_x\rangle$ is obtained from computational-basis measurement and $\langle\sigma_x\rangle$ requires a Hadamard rotation before measurement.

The VQE is swept over $\lambda \in [0, 1]$. Both the ground state and the excited staet acceced via $\theta_{\text{opt}} + \pi$ are computed. The VQE markers lie exactly on the exact diagonalization curves for both eigenvalues. Precision is shown in fig. 5 where the max error for the ground state is 5.41×10^{-11} and excited state is same order.

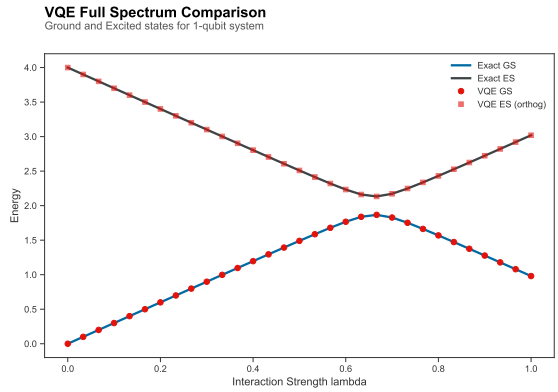


Figure 4. VQE Full Spectrum

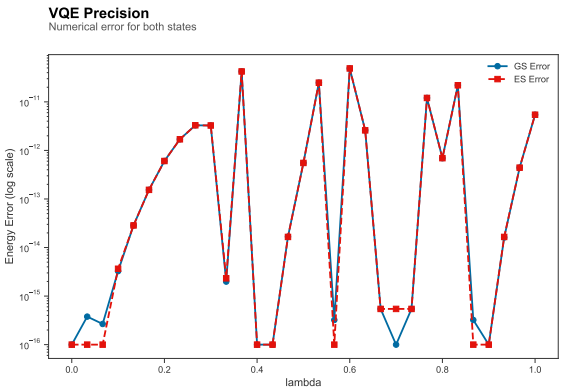


Figure 5. Caption

As the error remains below 10^{-10} across the entire λ range, this confirms that (i) the R_y ansatz is exact for this 1-qubit problem, (ii) the BFGS optimizer converges to the global minimum reliably, and (iii) the one-parameter landscape has no local minima.

An independent VQE-implementation using `Qiskit`'s `SparsePauliOp` and `Statevector` produces identical results, validating the manual implementation.

3.3 Two-Qubit System and Interaction-Driven Entanglement

The two-qubit Hamiltonian with parameters $H_x = 2.0, H_z = 3.0, \epsilon_{00} = 0.0, \epsilon_{10} = 2.5, \epsilon_{01} = 2.5, \epsilon_{11} = 7.0$ is diagonalized over $\lambda \in [0, 1]$.

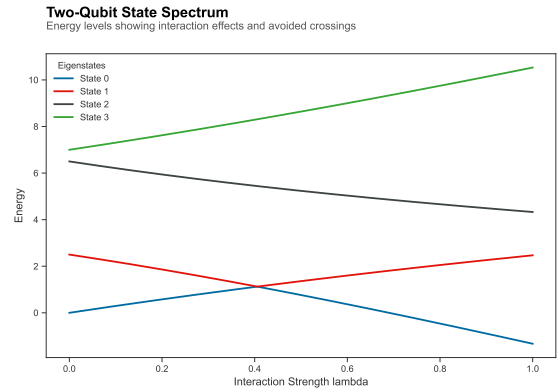


Figure 6. Eigenvalues

At $\lambda = 0$ the eigenvalues are the non-interacting energies $0.0, 2.5, 6.5, 7.0$. As λ increases, the $\sigma_x \otimes \sigma_x$ interaction shifts the diagonal elements, pushing $|00\rangle$ and $|11\rangle$ up while pulling $|01\rangle$ and $|10\rangle$ down. The $\sigma_x \otimes \sigma_x$ coupling creates off-diagonal mixing between $|00\rangle \leftrightarrow |11\rangle$ and $|01\rangle \leftrightarrow |10\rangle$ producing avoided crossings. the block structure of the Hamiltonian, two independent 2×2 blocks, are reflected in the pairwise behaviours of the energy levels.

The von Neumann entropy is computed for all four eigenstates, not just the ground state.

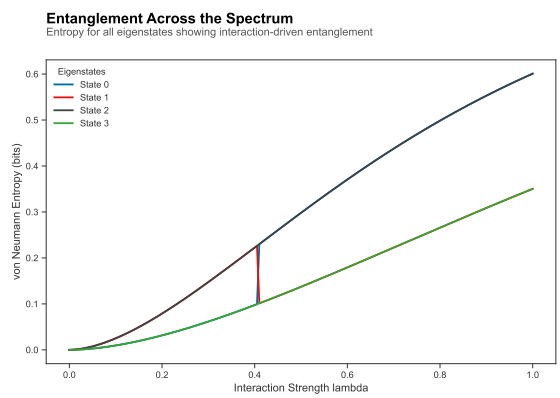


Figure 7. Entropy

Some key observations; (i) zero entropy, where at $\lambda = 0$ all eigenstates are computational basis states (product states), so $\mathcal{S} = 0$; (ii) monotonic increase, where as λ grows the off-diagonal interaction mixes the basis states, generating entanglement and the entropy increases smoothly; (iii) sharp jump at the avoided crossing, near the avoided crossing region, the entropy shows a rapid increase, the ground state transitions from being dominated by a single basis state to being strongly entangled superposition, a small change in λ produces a dramatic shift in quantum correlations, (iv) Hamiltonian-driven entanglement, the entanglement is not externally imposed, it emerges directly from the competition between the diagonal spacing and the off-diagonal coupling H_x, when the coupling becomes comparable to the energy gap, entanglement grows sharply.

The non-interacting Hamiltonian $\mathcal{H}_0 = \mathrm{diag}\,(\epsilon_{00}, \epsilon_{01}, \epsilon_{10}, \epsilon_{11})$ is decomposed into four Pauli strings, as seen in eq. (39), with $h_0 = 4.0, h_1 = 0.25, h_2 = -2.75, h_3 = -0.5$. The interaction adds $(h_3 + \lambda\mathcal{H}_z), ZZ + \lambda\mathcal{H}_x, XX$, giving 5 Pauli terms total.

$$\mathcal{H}_0 = h_0 \tag{39}$$
$$II + h_1 \tag{40}$$
$$ZI + h_2 \tag{41}$$
$$IZ + h_3 \tag{42}$$
$$ZZ \tag{43}$$

The hardware-efficient ansatz consists of two R_y rotation layers seperated by a CNOT entangler

$$|\psi(\theta)\rangle = (R_y(\theta_3) \otimes R_y(\theta_4)) \cdot \mathrm{CNOT} \cdot (R_y(\theta_1) \otimes R_y(\theta_2))\,|00\rangle \tag{44}$$

This gives 4 variational parameters. The R_y rotations span the real subspace of each qubit, and the CNOT provides the entanglement needed to represent correlated states. For a 4-dimensional real Hilbert space, 4 real parameters are sufficient.

The VQE sweep over $\lambda \in [0, 1]$ compares three approaches, in fig. 8, where all three curves, exact, VQE, and Qiskit VQE, are indistinguishable. With the precision, the max absolute error for manual VQE and Qiskit VQE are both $\approx 10^{-10}$.

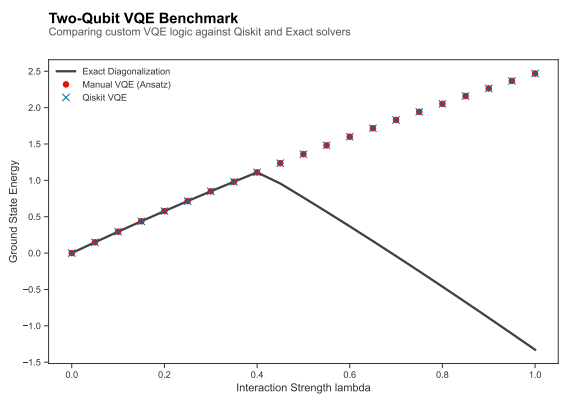


Figure 8. VQE Comparison

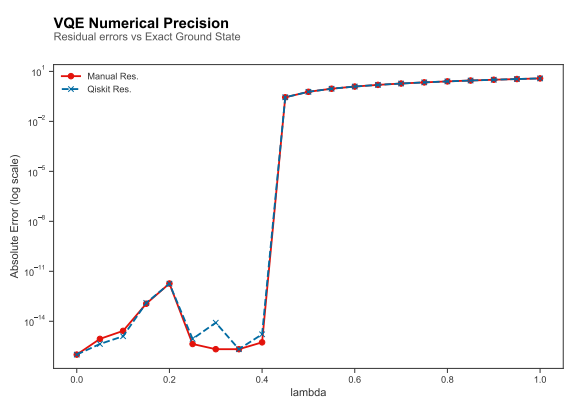


Figure 9. VQE Precision

The R_y-CNOT-R_y circuit can represent any real two-qubit state. This means the VQE achieves exact results, the error is purely from the optimizer, not the ansatz.

BFGS converges reliably from the zero-parameter initialization $\theta = (0, 0, 0, 0)$. No random restarts were needed, suggesting a smooth optimization landscape for this system.

The exact match between the manual implementation and Qiskit validates the VQE-code, both correctly handle the ansatz, Pauli measurements, and optimization loop.

The VQE implicitly captures the entanglement structure of the ground state, as λ increases through the crossing region, the optimal CNOT angle changes to produce the appropriate entanglement, tracked by the entropy analysis from the first part.

3.4 The Lipkin Model

The Lipkin Hamiltonian is built using the quasispin operators J_z, J_+, J_- with the pair-scattering interaction ($W = 0$), where the resulting matrices are shown for $J = 1$ with $N = 2$ particles and 3×3 in eq. (45) and $J = 2$ with $N = 4$ particles and 5×5 in eq. (46).

The $m = 0$ state is decoupled, it is an eigenstate with energy 0 for all V. The interaction only connects $m = -1 \leftrightarrow m = +1$, the $\Delta m = \pm 2$ selection rule from $J_+^2 + J_-^2$.

$$H_{J=1} = \begin{pmatrix} -\epsilon & 0 & -V \\ 0 & 0 & 0 \\ -V & 0 & \epsilon \end{pmatrix} \tag{45}$$

The chequerboard pattern $\Delta m = \pm 2$ persists, decoupling even-m and odd-m states into separate blocks.

$$H_{J=2} = \begin{pmatrix} -2\epsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\epsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \epsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\epsilon \end{pmatrix} \tag{46}$$

Mapping the quasispin operators to $N = 2$ qubits gives

$$H_{J=1} = \frac{\epsilon}{2}(ZI + IZ) + \frac{V}{2}(XX - YY) \tag{47}$$

For $J = 1$, two-qubit mapping. The 4×4 Pauli Hamiltonian contains both the $J = 1$ tripled $d = 3$ and $J = 0$ singlet $d = 1$. Projecting onto the triplet subspace, spanned by $|11\rangle$ ($m = -1$), $\frac{|01\rangle + |10\rangle}{\sqrt{2}}$ ($m = 0$) and $|00\rangle$ ($m = +1$) exactly covers the 3×3 quasispin matrix.

For $J = 2$, four-qubit mapping.

$$\boxed{H_{J=2} = \frac{\varepsilon}{2}\sum_{i=1}^{4} Z_i + \frac{V}{2}\sum_{i<j}(X_iX_j - Y_iY_j)}$$

This gives 4 single-qubit Z terms and 12 two-qubit terms (XX and YY for each of the 6 pairs), totalling 16 Pauli-strings. The 16×16 Hilbert space contains $J = 0, 1, 2$ sectors. The five $J = 2$ eigenvalues were extracted from the full Pauli spectrum and verified against the quasispin reference.

The eigenvalues are computed over $V \in [0, 1.5]$ with $\epsilon = 1$ and $W = 0$.

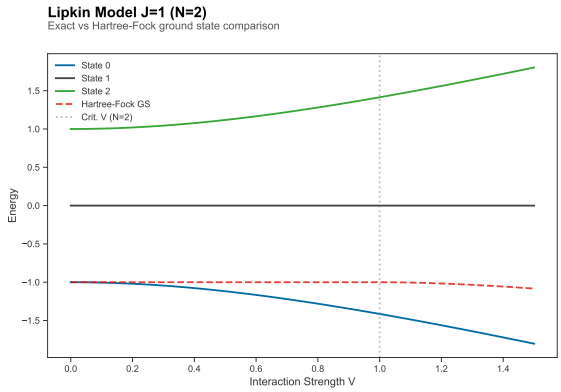


Figure 10. J1

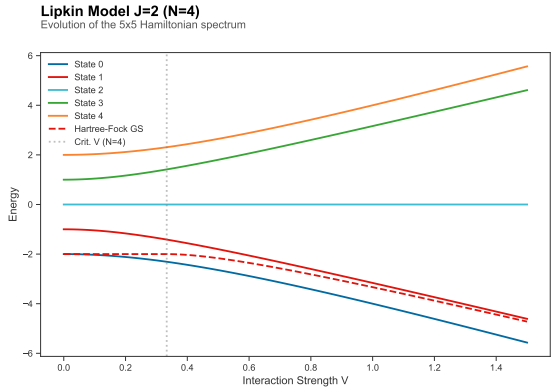


Figure 11. J2

Both spectra show the ground state energy decreasing with increasing V, reflecting energetically favorable pair correlations introduced by the interaction.

The Hartree-Fock energy of the Lipkin model is eq. (48) where $\tilde{V} = \frac{(N-1)V}{\epsilon}$, and the critical interaction strengths are for $N = 1(J = 1) = 1.000$ and $N = 4(J = 2) = 0.333$.

$$E_{\text{HF}} = \begin{cases} \frac{-N\epsilon}{2} & \text{if } \tilde{V} \leq 1 \\ \frac{-N\epsilon}{4}\left(\tilde{V} + \frac{1}{\tilde{V}}\right) & \text{if } \tilde{V} > 1 \end{cases} \tag{48}$$

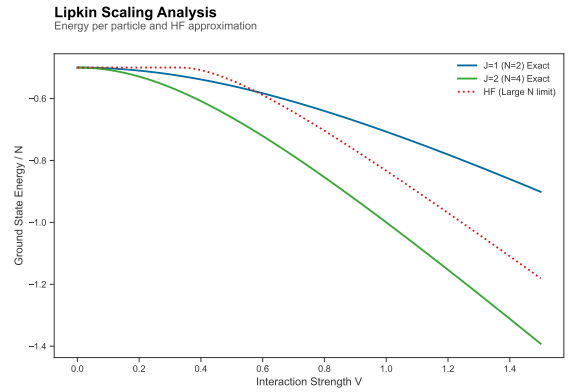


Figure 12. Scaling

The energy per particle $\frac{E_0}{N}$ approaches the Hartree-Fock curve as N increases, consistent with HF becoming exact in the thermodynamic limit. Below V_c HF gives the exact answer, the ground state is a single Slater determinant. Above V_c HF systematically overestimates the energy, the correlation energy is missing.

A hardware-efficient ansatz with R_y rotations and CZ entanglement is used and shown in table 5.

Sys	Qbts	Depth	Params	Restarts	Opt
$J = 1$	2	2	4	3	L-BFGS-B
$J = 2$	4	4	16	4	L-BFGS-B

Table 5. Variational quantum circuit configurations used for the Lipkin model simulations.

The ansatz uses alternating entanglement layers, even pairs $(1, 0), (2, 3)$ and odd pairs $(1, 2), (3, 4)$ plus wrap-around to ensure full qubit connectivity after two layers. Multiple random restarts mitigate local minima.

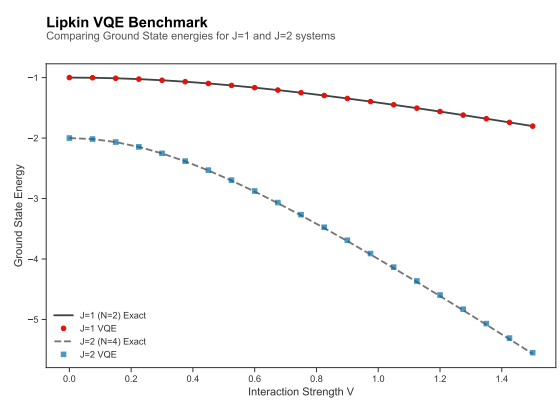


Figure 13. VQE Manybody

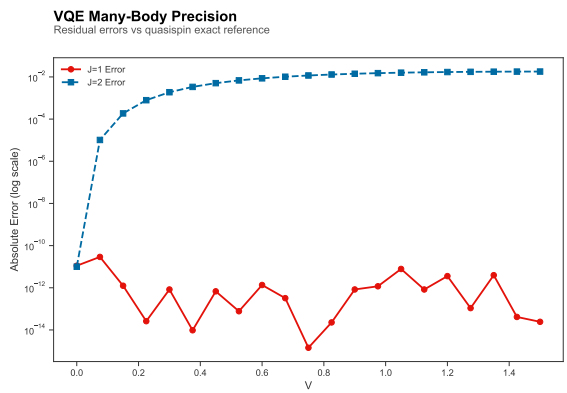


Figure 14. VQE Precision

With this we find a max error at $\approx 10^{-11}$ and $\approx 10^{-2}$ for $J = 1(N = 2)$ and $J = 2(N = 4)$ respectively, and with subsequent mean error of $\approx 10^{-12}$ and $\approx 10^{-3}$.

$J = 1$ exact accuracy.

$J = 2$ optimization challenges.

Possible improvements.

Phase transition captured.

Scaling.

4 Numerical Stability and Precision

5 Conclusion

5.1 Summary of Findings

5.2 Critique

5.3 Reflections

5.4 Future Work

References

Anthropic (2026). *Claude Large Language Model*. URL: https://www.anthropic.com.

Anysphere (2026). *Cursor AI Code Editor*. URL: https://cursor.sh.

Carleo, Giuseppe, Ignacio Cirac, Kyle Cranmer, Laurent Daudet, Maria Schuld, Naftali Tishby, Leslie Vogt-Maranto, and Lenka Zdeborová (2019). "Machine learning and the physical sciences". In: *Reviews of Modern Physics* 91, p. 045002. DOI: 10.1103/RevModPhys.91.045002.

Farhi, Edward, Jeffrey Goldstone, and Sam Gutmann (2014). "A Quantum Approximate Optimization Algorithm". In: arXiv: 1411.4028 [quant-ph].

Feynman, Richard P. (1982). "Simulating Physics with Computers". In: *International Journal of Theoretical Physics* 21.6–7, pp. 467–488. DOI: 10.1007/BF02650179.

Furnes, Egil (2026). *FYS5419 – Quantum Computing and Machine Learning Project*. GitHub repository containing source code and experiments. URL: https://github.com/egil10/fys5419.

GitHub (2026). *GitHub Copilot*. URL: https://github.com/features/copilot.

Google (2026). *Antigravity: AI-powered development environment for Gemini*. AI-assisted development environment built on Google Gemini models. URL: https://antigravity.dev.

Google DeepMind (2026). *Gemini Large Language Model*. URL: https://deepmind.google.

Hjorth-Jensen, Morten et al. (2026). *QuantumComputingMachineLearning*. GitHub repository. URL: https://github.com/CompPhysics/QuantumComputingMachineLearning.

IBM Quantum (2024). *Qiskit: An Open-source Framework for Quantum Computing*. URL: https://qiskit.org.

Lipkin, H. J., N. Meshkov, and A. J. Glick (1965). "Validity of many-body approximation methods for a solvable model: (I). Exact solutions and perturbation theory". In: *Nuclear Physics* 62, pp. 188–198. DOI: 10.1016/0029-5582(65)90862-X.

McClean, Jarrod R., Jonathan Romero, Ryan Babbush, and Alán Aspuru-Guzik (2016). "The theory of variational hybrid quantum-classical algorithms". In: *New Journal of Physics* 18.2, p. 023023. DOI: 10.1088/1367-2630/18/2/023023.

NumPy Developers (2024). *NumPy: Fundamental package for scientific computing with Python*. URL: https://numpy.org.

OpenAI (2023). *ChatGPT*. Large language model. URL: https://chat.openai.com.

Peruzzo, Alberto, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O'Brien (2014). "A variational eigenvalue solver on a quantum processor". In: *Nature Communications* 5, p. 4213. DOI: 10.1038/ncomms5213.

Schuld, Maria and Nathan Killoran (2021). *Machine Learning with Quantum Computers*. Cham: Springer. DOI: 10.1007/978-3-030-83098-4.

Schuld, Maria and Francesco Petruccione (2018). *Supervised Learning with Quantum Computers*. Cham: Springer. DOI: 10.1007/978-3-319-96424-9.

xAI (2026). *Grok Large Language Model*. URL: https://x.ai.

Xanadu AI (2024). *PennyLane: Automatic differentiation of hybrid quantum-classical computations*. URL: https://pennylane.ai.

6 Appendix

Part A: Single-qubit gates and measurements

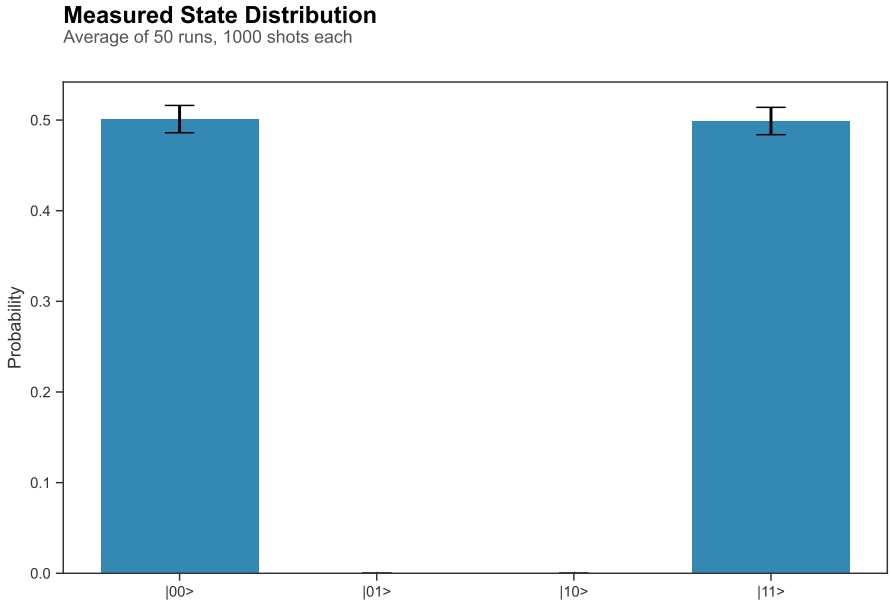
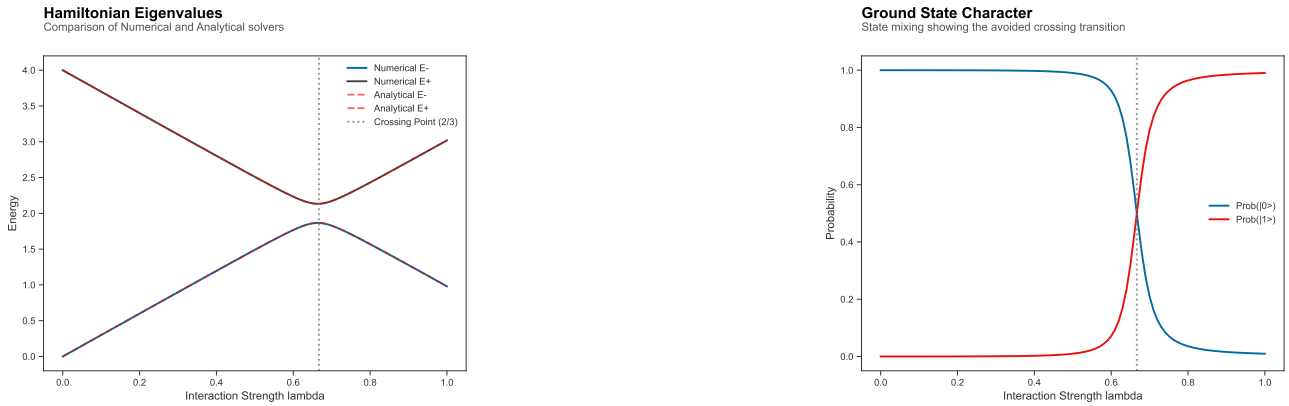


Figure 15. Measurement outcome distribution.

Part B: Two-level model (exact diagonalization)

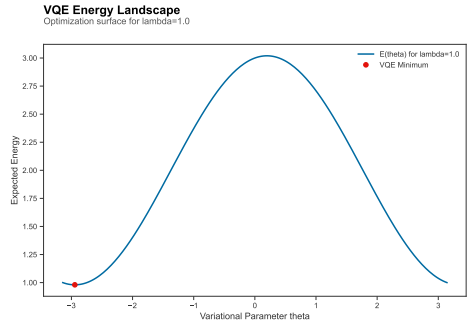


(a) Eigenvalues vs. λ.

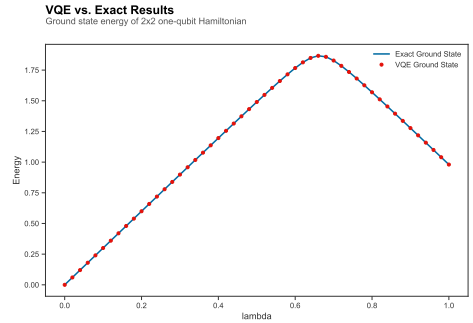
(b) Eigenvector mixing.

Figure 16. Exact spectrum and state character for the 2×2 Hamiltonian.

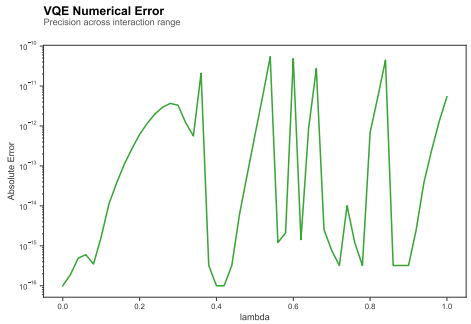
Part C: VQE for the two-level model



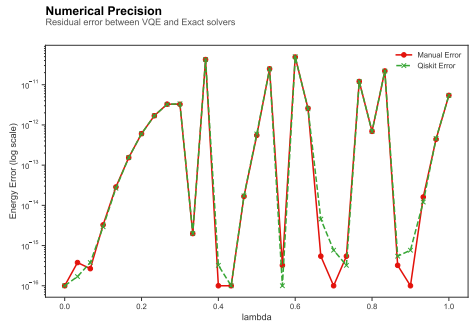
(a) Energy landscape.



(b) VQE vs. exact.

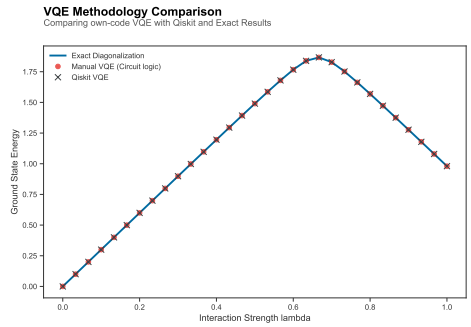


(c) Absolute error.

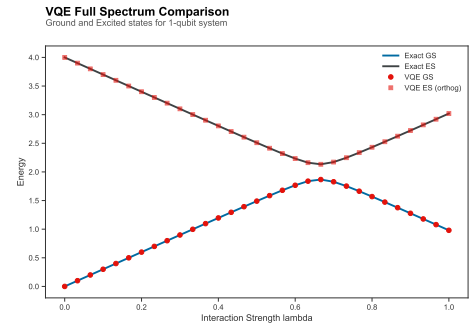


(d) Precision.

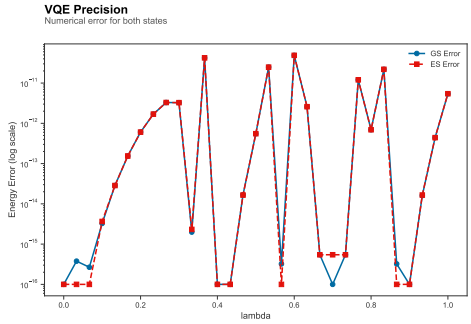
Figure 17. VQE optimization diagnostics for the 2×2 model.



(a) Method comparison.



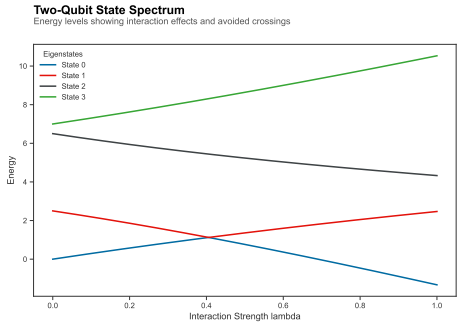
(b) Full spectrum.



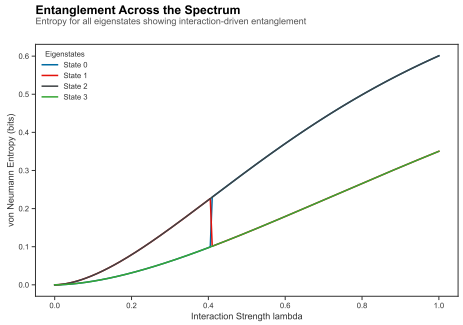
(c) Spectrum error.

Figure 18. Spectrum-level VQE comparisons for the 2×2 model.

Part D: Two-qubit model and entanglement



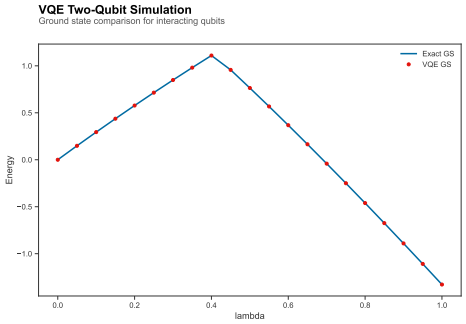
(a) Eigenvalues vs. λ.



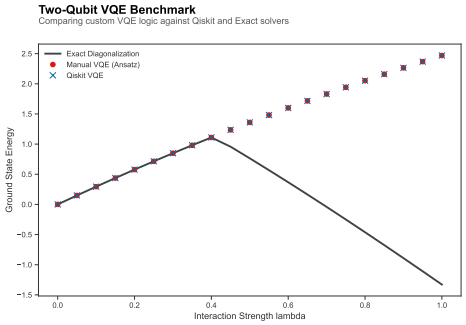
(b) Von Neumann entropy.

Figure 19. Spectrum and entanglement diagnostics for the two-qubit Hamiltonian.

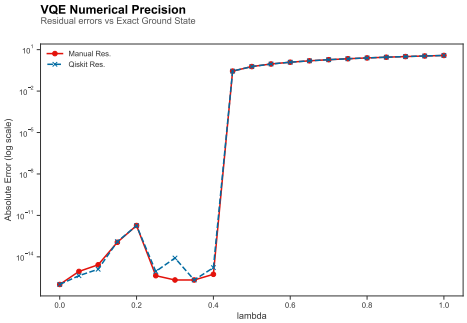
Part E: VQE for the two-qubit model



(a) VQE vs. exact.



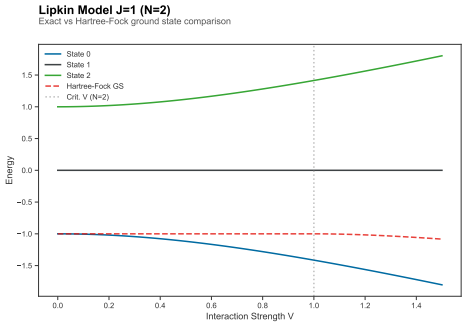
(b) Comparison.



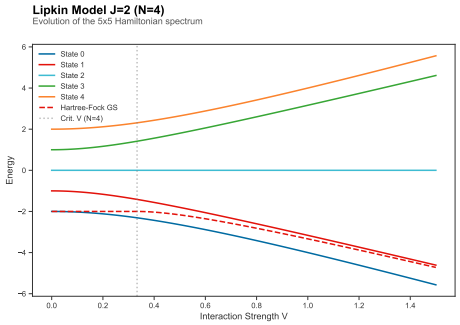
(c) Precision.

Figure 20. VQE performance for the two-qubit Hamiltonian.

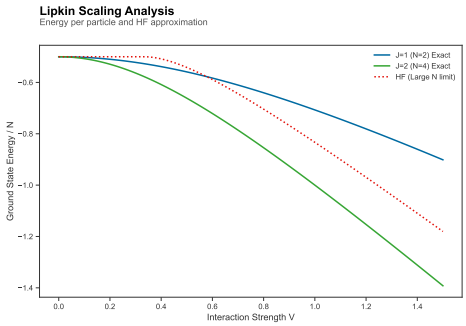
Part F: Lipkin model (classical reference)



(a) $J = 1$.



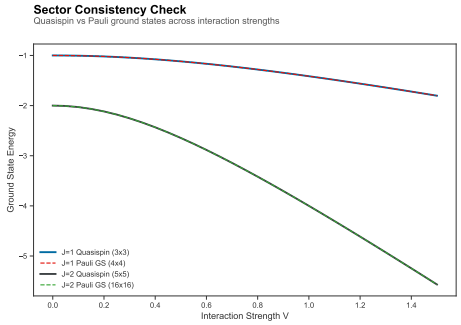
(b) $J = 2$.



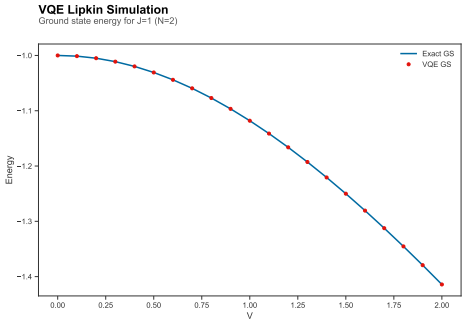
(c) Scaling.

Figure 21. Classical diagonalization results for the Lipkin model.

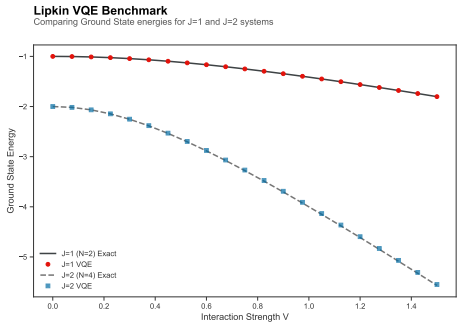
Part G: Lipkin model with VQE



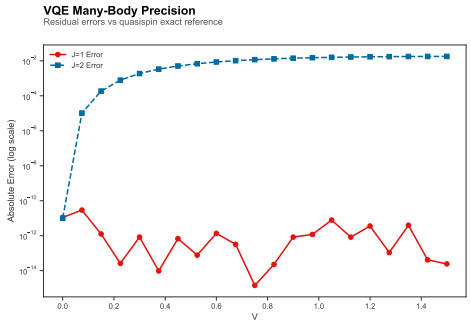
(a) Sector check.



(b) VQE ($J = 1$).



(c) Many-body VQE.



(d) Precision.

Figure 22. VQE diagnostics for the Lipkin model.

6.1 Programming with `python`

```
absl-py>=2.1.0
cplex>=22.1.2.0
gurobipy>=13.0.1
hmmlearn>=0.3.3
matplotlib>=3.7.1
nautilus_trader>=1.155.0
numpy>=1.24.3
pandas>=2.2.2
pennylane>=0.38.0
pyarrow>=17.0.0
pylatexenc>=2.10
qiskit>=1.0.0
qiskit-aer>=0.17.2
qiskit-algorithms>=0.4.0
qiskit-ibm-runtime>=0.45.1
qiskit-optimization>=0.7.0
scipy>=1.14.1
seaborn>=0.13.2
sympy>=1.13.3
vectorbt>=0.26.0
```

Listing 2. Python environment and core libraries used for the numerical simulations, classical diagonalization, and quantum-circuit implementations in this project. The environment includes scientific computing tools (NumPy, SciPy, pandas), visualization libraries (Matplotlib, Seaborn), and quantum computing frameworks (PennyLane and Qiskit).